

LiDAR-Based Lane Marking Segmentation in Urban Point Clouds Using RandLA-Net

by

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Bachelor Thesis

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Abstract

Consider this a separate document, although it is submitted together with the rest. The abstract aims at another audience than the rest of the proposal. It is directed at the final decision maker or generalist, who typically is not an expert at all in your field, but more a manager kind of person. Thus, don't go into any technical description in the abstract, but use it to motivate the work and to highlight the importance of your project.

(target size: 15-20 lines)

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1 Introduction

This, like the rest, addresses fellow experts from your field (but not from your particular topic of research). Here you should technically connect to the main concepts from that field and give an outline of your project, stating the research/engineering question that you want to get answered by your project.

(target size: 1-2 pages)

2 Statement and Motivation of Research

This part should make clear which question, exactly, you are pursuing, and why your project is relevant/interesting. This is the place to explain the background and to review the existing literature. Where does your project extend the state of the art? What weaknesses in known approaches do you hope to overcome? If you have carried out preliminary experiments, describe them here.

(target size: 5-10 pages)

3 Description of the Investigation

This is the technical core of the thesis. Here you lay out your how you answered your research question, you specify your design of experiments or simulations, point out difficulties that you encountered, etc.

(target size: 5-10 pages)

4 Evaluation of the Investigation

This section discusses criteria that are used to evaluate the research results. Make sure your results can be used to published research results, i.e., to the already known state-of-the-art.

(target size: 5-10 pages)

5 Conclusions

Summarize the main aspects and results of the research project. Provide an answer to the research questions stated earlier.

(target size: 1/2 page)

A Typesetting Examples

A.1 Code Fragments

```
fn num_to_ordinal_expr(x: u32) -> String {  
  format!("{}", x, match (x % 10, x % 100) {  
    (1, 1) | (1, 21...91) => "st",  
    (2, 2) | (2, 22...92) => "nd",  
    (3, 3) | (3, 23...93) => "rd",  
    _ => "th"  
  })  
}
```

A.2 Tables

Number	Description
7	A lucky number in Western culture
8	A lucky number in Chinese and other Asian cultures
42	Answer to the ultimate question of life, the universe, and everything
404	Not found

Table 1: Useless insights I gained with no further meaning

A.3 Plots

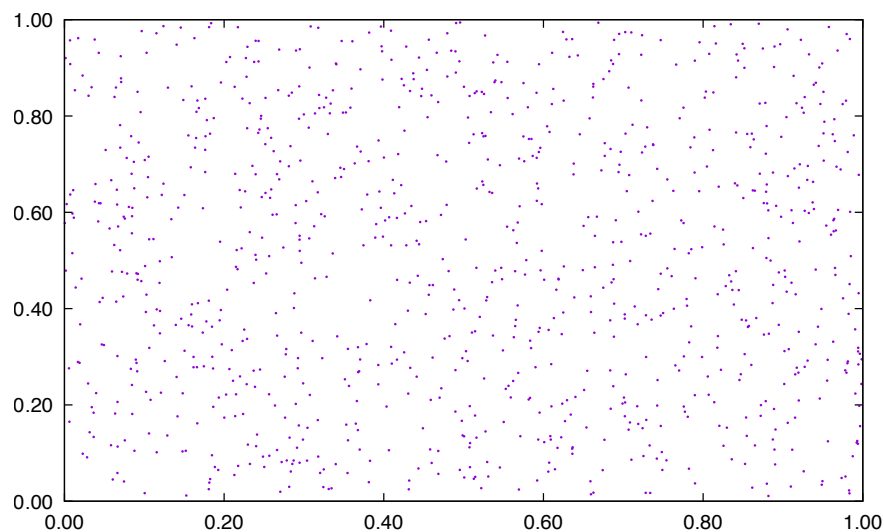


Figure 1: Many dots distributed over a two dimensional unit space without any discernible pattern or deeper meaning